

Arithmetic and complexity

Explicit tensors with quadratic border rank

Difficulty: extreme **Importance:** broadly interesting**Status:** Open

Rating rationale: Extreme because explicit superlinear tensor lower bounds remain far below generic quadratic behavior; broad importance is the construction of hard instances for bilinear computation and circuit lower bounds.

Topic: explicit lower bounds for tensor decompositions

Context and notation

Over $\mathbb{C}$, the tensor rank $R(T)$ is the minimum number of pure tensors in an exact sum for T . The border rank $\underline{R}_{\mathbb{C}}(T)$ is the least r such that T is a Euclidean limit of complex tensors of rank at most r .

Problem statement

Construct a deterministic algorithm that, on input the positive integer n , outputs the rational entries of a tensor $T_n \in \mathbb{Q}^{n \times n \times n}$ in time polynomial in n , and prove that there are constants $c > 0, n_0$ such that $\underline{R}_{\mathbb{C}}(T_n) \geq cn^2$ for all $n \geq n_0$. Output rationals are encoded by binary integer numerators and denominators; thus the time bound also limits their bit lengths. Border rank is over $\mathbb{C}$, as defined in the notation.

Why it matters

Explicit hard tensors would supply lower bounds for bilinear computation that are much stronger than the currently available examples. Generic existence does not provide the required efficient construction.

References and status

M. Michałek and J. Landsberg, *Towards Finding Hay in a Haystack: Explicit Tensors of Border Rank Greater Than $2.02n$* , Theory of Computing 21(13), 2025, §1 and Definition 1.1, distinguishes polynomial-time explicitness from weaker notions and obtains a linear bound. The quadratic target and rational-output encoding are an editorial quantitative formulation of the paper’s explicit-versus-generic challenge, not a numbered conjecture quoted from it. Searches for “explicit tensors superlinear rank 2026” and “explicit tensors superlinear border rank” located no quadratic construction; weaker semi-explicit constructions do not meet this statement. **Admitted: no resolution located.**

Status check — 2026-09-10

Rechecked the 2025 Landsberg–Michałek paper, and searched for explicit superlinear or quadratic border-rank constructions. The July 2026 Arithmetic circuit lower bounds from sumset expansion, also ECCC TR26-138, gives semi-explicit tensors over number fields. That does not meet the displayed polynomial-time rational-output requirement. No qualifying quadratic construction was located.

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Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. [Verified affiliation and submission record.](#)

Theorems 2.1–5.3 prove compaction, conditional reductions and a scoped method ceiling. Section 6 supplies finite certificates for dimensions 2–9. A polynomial-time detecting family and all-dimension quadratic lower bound remain missing.

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